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CODA: Cascaded Online Discontinuity-Aware Alignment for Real-Time Score Following

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Summary

The paper presents CODA, an image-based real-time score following system that reframes tracking as cascaded selection over known score candidates rather than repeated page-wide detection. The model first selects the active system, then the bar within that system, and finally regresses a bar-local note position, enforcing geometric consistency and reducing search space. To handle repeats and arbitrary score jumps online, CODA introduces a silence-driven “break mode” and complementary jump augmentation during training. On the MSMD piano benchmarks, CODA attains state-of-the-art accuracy under real-time constraints and shows strong recovery on a new repeat-aware jump evaluation the authors contribute.

Strengths

Technical novelty and innovation

- Cascaded selection formulation (system → bar → note) that exploits known layout candidates and enforces geometric consistency across levels, directly addressing instability of multi-head detection approaches.
- Audio-conditioned selection with causal Mamba encoder, FiLM modulation, and cross-attention over recent audio states—an elegant multimodal design tailored to streaming.
- Beam search with learned, bounded transition penalties as soft temporal priors; practical and principled for online decoding without hard monotonicity.
- A simple, effective discontinuity mechanism: energy-triggered "break mode" that relaxes priors and widens beams, paired with jump augmentation to close the train–test gap.

Experimental rigor and validation

- Standardized comparisons on MSMD against established image-based baselines under identical protocols, showing consistent improvements across thresholds and at system/bar levels.
- Ablation study quantifying contributions of cascade, cross-attention, beam search, temporal priors, and scheduled sampling.
- A dedicated jump benchmark with clear metrics (recovery rate, latency, post-jump accuracy) and analysis of repeat vs random jumps.

Clarity of presentation

- Clear problem re-formulation and factorization, with a readable architectural narrative and explicit decoding and training procedures.
- Well explained jump handling with diagrams and parameter settings; practical implementation details and runtime reported.

Significance of contributions

- Advances the state of the art for real-time image-based score following, a problem of recurring interest for interactive music systems.
- Introduces a principled approach to discontinuities in the image-based setting, an acknowledged gap in prior work.
- Contributes a repeat-aware jump evaluation protocol that can catalyze more rigorous research on jumps in image-based tracking.

Weaknesses

Technical limitations or concerns

- Assumes accurate system/bar bounding boxes from layout metadata; reliance on this supervision limits immediate applicability to scanned scores without robust OMR or layout detection.
- Break-mode detection is based on simple energy thresholds; robustness under noisy environments, soft/legato playing, or pedal-sustained textures may be brittle.
- The learned "page-local" transition priors are insufficiently specified; it is unclear whether they are global parameters, conditioned on layout, or per-page embeddings (raising potential generalization and leakage concerns).

Experimental gaps or methodological issues

- All main experiments are on MSMD (synthetic images and mostly synthetic audio), with limited real-audio testing and no scanned-score evaluations (Settings III/IV from prior work are excluded due to availability).
- Cross-modality fairness: CODA benefits from temporal beam decoding and priors, while the strongest baseline (CYOLO-SB) is framewise; it would strengthen claims to show a smoothed/temporal-decoding baseline for CYOLO-SB.
- Jump evaluation appears conducted on synthetic audio from MSMD; no repeat-handling results are shown on real recordings, where silence detection and relocalization may be harder.

Clarity or presentation issues

- Details of the transition prior parameterization are ambiguous ("page-local" vs "global"), and how these priors scale with variable numbers of systems/bars needs clarification.
- The note-localization head regresses a continuous bar-local point for polyphonic piano without discussion of handling multi-voice simultaneity or ambiguous mid-bar alignments.

Missing related work or comparisons

- While symbolic followers are cited, a brief quantitative or qualitative contrast to recent low-latency "transcribe-and-track" symbolic pipelines would help position image-based trade-offs.

- Some recent offline image–audio alignment efforts with explicit repeat handling are cited but not used as reference points for jump recovery quality (even acknowledging offline vs online differences).

Detailed Comments

Technical soundness evaluation

- The cascaded factorization p(s|h)p(b|s,h)p(u|b,s,h) is well motivated and practically compelling in static page settings; reducing search to known candidates is an important insight relative to detection-based schemes. Using ROI Align over shared visual features with audio-conditioned modulation and cross-attention is technically sound and matches the streaming constraint (Mamba provides causal sequence modeling with efficient inference).
- Beam search with bounded non-positive transition costs is a pragmatic way to encode smoothness while remaining interruptible by strong audio evidence; clamping stay=0 is sensible to avoid degenerate decay. However, the formulation of "page-local" priors remains under-specified: if priors are parameterized per possible step size or adjacency class, that should be explicit. If they are per-page learned parameters, clarify how they are trained without leaking test metadata and how they generalize to unseen pages.
- The scheduled sampling-like curriculum for cascading is appropriate and addresses exposure bias; masking downstream losses when the selected system is wrong is a reasonable compromise to avoid supervising impossible targets while still correcting upstream predictions.
- The silence-triggered jump mechanism is simple and coherent with empirical observations (breaks often precede jumps). It is also general across repeats, D.C., coda, and performer errors. That said, silence-based triggers are vulnerable where performers do not pause or where room noise masks silence, so the approach should be viewed as a strong baseline rather than a complete structural model.

Experimental evaluation assessment

- The main tables show robust improvements over strong image-based baselines under the exact same MSMD protocol, including multi-

resolution metrics. Gains are sizeable (.914 vs .837 at ≤0.10 s; bar acc .975 vs .890), suggesting the cascade and decoding provide real benefits. The ablation study supports the central design choices, especially the cascade.

- Real-audio results (Setting II) retain sizable gains, which is encouraging for generalization. However, broader real-world generalization (scanned scores, diverse mic/room conditions) remains untested; this gap is acknowledged in the discussion.
- The jump benchmark and metrics are thoughtful; the analysis separating repeat vs random jumps is informative. The comparison to "CODA w/o break" and CYOLO-SB highlights that both a trigger and relaxed priors are necessary for fast recovery. Still, it would be valuable to include results on real recordings with repeats to demonstrate robustness under realistic acoustics.
- Runtime claims (12.8 ms per 20-fps frame) are appropriate for real-time use and reflect the efficiency of the proposed pipeline and pruning strategy.

Comparison with related work (using the summaries provided)

- Relative to CYOLO(-SB) [Henkel & Widmer], CODA avoids per-frame detection and exploits known score structure, which the results validate. Compared to symbolic followers (e.g., Nakamura et al., Matchmaker, and "transcribe-and-track" pipelines), CODA removes the need for symbolic scores but inherits the challenges of visual layout and image/audio modality gaps. The proposed jump recovery echoes the "break state" insight of Nakamura et al. but adapts it to the image-based regime via energy-triggered decoding relaxations rather than explicit HMM transitions.
- Offline repeat-aware synchronization (Shan & Tsai) and recent human-in-the-loop repeat labeling pipelines have shown strong repeat/jump handling offline; CODA's contribution is to bring a practical, generalizable jump mechanism to the real-time, image-based setting.
- Recent unified or symbolic approaches (e.g., RUMAA; transcribe-and-track) report excellent alignment with explicit repeat modeling but are not real-time image-based systems; CODA fills this complementary space with a lightweight, low-latency multimodal approach.

Discussion of broader impact and significance

- The cascaded selection reframing is likely to influence future image-based score following by decoupling localization from unnecessary

detection and better reflecting the fixed nature of score layouts. The jump benchmark provides much-needed evaluation scaffolding for discontinuity handling. Practically, CODA's speed and accuracy make it relevant for accompaniment, page turning, and education tools.

- Limitations—dependence on layout metadata, sensitivity of energy-based break detection, and narrow training domain—are important; addressing them will be key to deployment on scanned, noisy, or multi-instrument scenarios. The approach appears extensible: integrating robust measure/system detection, adaptive silence thresholds, or lightweight structural priors could broaden applicability.

Questions for Authors

1. How exactly are the "page-local transition penalties" parameterized and learned? Are they global parameters shared across pages, functions of step size (e.g., Δbar), or per-page embeddings? How do you avoid test-time leakage if they are page-specific?
2. What is the source of the system/bar bounding boxes at test time? On MSMD they are available; for scanned scores, do you envision an upstream detector, and how robust is CODA to moderate box errors or missing bars?
3. How sensitive is break-mode performance to the energy thresholds and hysteresis settings under real recording conditions (e.g., background noise, sustain pedal)? Did you test adaptive or learned silence detection (e.g., VAD) and if so, with what effect?
4. In the note head, you regress a single bar-local coordinate u for polyphonic piano. How do you interpret this target when multiple onsets occur simultaneously? Is the supervision tied to the earliest onset in the bar segment, the note of interest at the evaluation time, or a geometric proxy?
5. Can you report jump-recovery results on the real-audio subset with scores containing repeats? If not available, could you at least discuss expected failure modes and whether energy-based break detection remains reliable?
6. How often does the beam search prune away the correct hypothesis (k=3, m=3), and how does increasing k,m affect latency and accuracy? A sensitivity curve would be useful.

7. Have you compared CODA to a temporally smoothed version of CYOLO-SB (e.g., with simple HMM/beam decoding over its system/bar heads) to isolate the benefit of cascaded selection vs decoding?
8. Does cross-attention over the Mamba buffer materially change results compared to a simpler pooling of recent audio (e.g., GRU last state or mean pooling)? Any additional ablation?
9. How does CODA behave when performers do not pause before a jump (e.g., immediate D.C. without silence)? Does the system eventually recover, and with what latency, if break mode is never triggered?

Overall Assessment

This paper makes a strong, well-argued contribution to real-time image-based score following. The cascaded selection formulation is both conceptually clean and empirically effective, yielding substantial accuracy gains and stability improvements over prior detection-style approaches. The silence-driven break mode and training augmentation provide a practical, modality-agnostic mechanism to recover from repeats and arbitrary jumps—an underserved area in image-based tracking. The experimental results are convincing within the MSMD regime, and the ablations support the design choices. The main limitations are scope and robustness: dependence on perfect layout metadata, a relatively narrow evaluation domain (synthetic images; limited real audio; no scanned scores), and a simple energy-based jump trigger whose generality under real conditions is unproven. These concerns are acknowledged and, in my view, do not diminish the core methodological advances. I recommend acceptance, with encouragement to clarify the transition priors, extend jump-recovery tests to real recordings, and discuss integration with robust layout detection for scanned scores.

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